

Symbolic Explainable Artificial Intelligence

Programming Project: Problem as a CSP

Academic Year 2026/27

12/09/2026

Overview

The objective of the Symbolic Explainable Artificial Intelligence project is to develop an encoding for solving a problem as a Constraint Satisfaction Problem (CSP). The problem is to be chosen by each group of students from CSPLib (<https://www.csplib.org/Problems/>).

Each group is required to solve a **distinct problem**. For this reason, students are advised to choose a problem as soon as possible. Alternatively, groups may work on a problem not listed in CSPLib, provided that they receive prior approval from faculty, which must be requested by email until the 21st of September.

A shared file for the selection of problems by groups of students will be provided.

Project Goals

You are implementing a program in Python in the file `proj.py`. The figure below illustrates the workflow

Depending on the problem, you may need to take as input the information to characterize the specific problem instance to be solved, or even the name of a file representing the input data. For each specific problem instance, your program should write the solution to the standard output.

This program should use the MiniZinc CSP solver (recommended version 2.10.1 available from <https://www.minizinc.org/>). Alternatively, you may consider using MiniZinc Python (<https://python.minizinc.dev/>).

Figure 1 illustrates the expected pipeline for `proj.py`: it first reads the problem instance, encodes it as a MiniZinc model, invokes the solver, and finally parses the solver's output to produce the final solution on standard output.

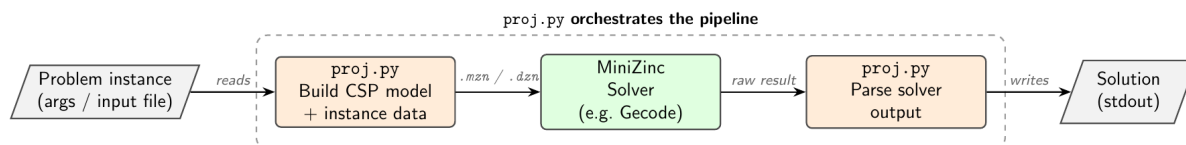


Figure 1: Expected workflow of `proj.py`.

Very Important: If you choose a problem for which there is already a MiniZinc model in CSPLib, you must include a non-trivial extension to the problem in your project. You may use the CSPLib model as a starting point.

Additional Information

1. Group Size & Language

- The project is to be implemented in groups of **one or two students**. Registration through Fenix is required.
- The recommended programming language to use is **Python version 3.12**.

Submission

The project code and the description of the work are to be submitted through the course website. You should submit a zip archive named `<group_number>.zip` containing:

- The source code of your solution.
- A **1-page PDF**¹ describing the work, including:
 - The CSPLib problem number (or details of the self-proposed problem).
 - The extension of the CSPLib problem that has been implemented (if applicable).
 - The URL link to your video submission (e.g., YouTube, Google Drive).
 - Instructions on how to run your code, including software versions.
 - A clear explanation of which web solutions or Generative AI tools were used in your work and how their contributions were integrated in your final project. This reflection is mandatory.

Video Requirements

- The video should have a maximum duration of **3 + 2 minutes**.
- All members of the group must appear on camera.
- **Structure:**
 - *First 3 minutes:* Explanation of the work developed.
 - *Last 2 minutes:* Demonstration of the developed software.

Evaluation Criteria

The evaluation will take into account the following:

- **Quality of the solution:** 50%
- **Video presentation:** 50%

If necessary, an oral evaluation for specific groups may be performed.

¹Ideally one page, two pages at most.

Project Plagiarism Detection & AI Tools

The same or very similar projects will lead to failure in the course and, eventually, to disciplinary proceedings. Your programs will also be compared with solutions from the web.

In this project you are allowed the use of Generative AI tools at the Yellow Level -Nível 1 (Amarelo)-, according to the rules in [“Guia para a Utilização Responsável da Inteligência Artificial”](#). This means that you are allowed a limited use of resources and you are required to add a declaration clarifying what tools or resources you used and how they were used. (As specified in section Submission.)

If you use AI-based tools or existing web solutions, you must explicitly describe their use in the video.

Project Dates

- **Project published:** 12/09/2026
- **Fenix group registration:** 25/09/2026
- **Project due:** 16/10/2026 (Zip file due at 17h00; Video due at 23h59)

Omissions & Errors

Any required clarifications will be made available through the course’s official website.